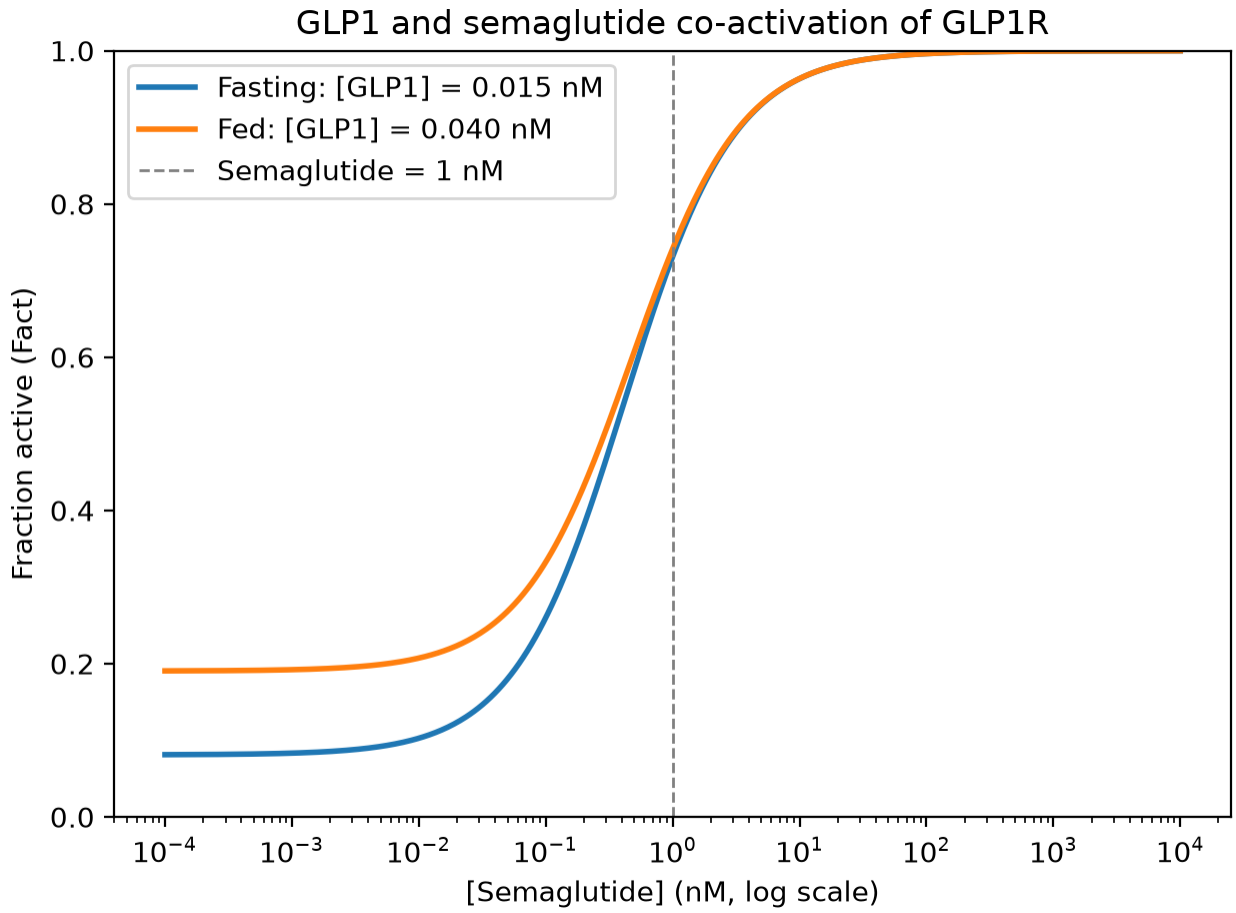
Module 3: GLP1 and semaglutide

Concentration response of GLP1R



Interpretation

Increasing semaglutide raises GLP1R activation toward 100% in both conditions, with fed GLP1 producing greater activation than fasting GLP1 (19.05% vs. 8.11% without semaglutide, and 74.14% vs. 73.12% at 1 nM semaglutide), so the difference between the curves shrinks as the receptor population saturates.

Model assumptions

At equilibrium, GLP1 and semaglutide compete for one receptor binding site, either bound agonist fully activates the receptor, and free ligand concentrations remain fixed. The plotted response is the total fraction of receptors occupied by either agonist.

GLP1_GLP1R.py

```
# GLP1_GLP1R.py

import numpy as np
import matplotlib.pyplot as plt

def Fact(GLP1, S, Kd_GLP1, Kd_S):
    """Fraction of GLP1R activated by either GLP1 or semaglutide."""
    # Equilibrium binding: the two agonists compete for one binding site.
    # Either bound agonist activates the receptor; free ligand stays fixed.
    # Total receptor = free receptor + GLP1-bound + semaglutide-bound.
    return (GLP1 / Kd_GLP1 + S / Kd_S) / (1 + GLP1 / Kd_GLP1 + S / Kd_S)

Kd_GLP1 = 0.17                # GLP1 dissociation constant [nM]
Kd_S = 0.38                   # semaglutide dissociation constant [nM]
GLP1_fasting = 0.015          # fasting GLP1 [nM]
GLP1_fed = 0.040              # GLP1 after a meal [nM]
S = np.logspace(-4, 4, 500)   # semaglutide concentration [nM]

plt.figure()
plt.semilogx(S, Fact(GLP1_fasting, S, Kd_GLP1, Kd_S), lw=2,
             label="Fasting: [GLP1] = 0.015 nM")
plt.semilogx(S, Fact(GLP1_fed, S, Kd_GLP1, Kd_S), lw=2,
             label="Fed: [GLP1] = 0.040 nM")
plt.axvline(1, ls="--", c="gray", lw=1, label="Semaglutide = 1 nM")
plt.ylim(0, 1)
plt.xlabel("[Semaglutide] (nM, log scale)")
plt.ylabel("Fraction active (Fact)")
plt.title("GLP1 and semaglutide co-activation of GLP1R")
plt.legend()
plt.tight_layout()

# Calculate the baseline separately because zero is not on a log axis.
print("GLP1 alone (no semaglutide):")
print("Fasting:", round(100 * Fact(GLP1_fasting, 0, Kd_GLP1, Kd_S), 2), "%")
print("Fed:", round(100 * Fact(GLP1_fed, 0, Kd_GLP1, Kd_S), 2), "%")
print("With 1 nM semaglutide:")
print("Fasting:", round(100 * Fact(GLP1_fasting, 1, Kd_GLP1, Kd_S), 2), "%")
print("Fed:", round(100 * Fact(GLP1_fed, 1, Kd_GLP1, Kd_S), 2), "%")

# Interpretation: Increasing semaglutide raises GLP1R activation toward 100%
# in both conditions, with fed GLP1 producing greater activation than fasting
# GLP1 (19.05% vs. 8.11% without semaglutide, and 74.14% vs. 73.12% at 1 nM
# semaglutide), so the difference between the curves shrinks as the receptor
# population saturates.

plt.show()
```